Part-Aware Policy Learning for Generalizable Robotic Manipulation

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Motivation and Background

Robotic manipulation policies often rely on global visual observations, which can make them sensitive to changes in object geometry, appearance, or configuration. However, many manipulation tasks depend mainly on interacting with a specific functional part, such as a drawer or cabinet handle. Explicitly representing the task-relevant part may therefore help a policy generalize across object instances. This project investigates whether part-aware visual representations improve manipulation robustness and generalization compared with policies using only global observations.

Problem Definition

We consider single-object manipulation tasks in simulation, such as opening an articulated drawer or cabinet. The task and target object are fixed, allowing us to isolate the effect of part-level information without introducing language grounding.

The policy receives an RGB observation and robot proprioceptive state and predicts continuous robot actions. The proposed part-aware policy additionally receives a representation of the task-relevant object part derived from simulator ground-truth segmentation. The primary evaluation metric is task success rate on both seen and unseen object instances, with additional tests under variations in part location, object geometry, and initial configuration.

Proposed Direction

We will compare two imitation-learning policies:

- **Baseline:** global visual features + robot state $\rightarrow$ robot action.
- **Part-aware:** global visual features + target-part features + robot state $\rightarrow$ robot action.

A pretrained visual encoder such as DINOv2 will extract image features. For the proposed model, the target-part mask will be used to pool features from the manipulation-relevant region. These features will be combined with robot state and passed to a behavior-cloning policy, initially using an MLP and potentially an Action Chunking Transformer (ACT). This design isolates whether explicit part information itself improves manipulation generalization. As a future extension, language instructions can be added to enable target object and target-part selection in multi-object environments.

Robot, Simulation, and Data

We will use **ManiSkill3** with articulated manipulation tasks involving drawers or cabinets. Demonstration trajectories will provide RGB observations, robot states, and expert actions. During trajectory replay or data collection, simulator segmentation and object/link information will be used to obtain ground-truth target-part masks.

Training and test sets will use different object instances when possible so that the policy can be evaluated on unseen geometries and part configurations. Additional perturbations, such as changes in object pose or robot initialization, may be introduced for robustness evaluation.

Target Resources

The project will use ManiSkill3 environments and demonstrations, pretrained visual encoders such as DINOv2, and existing imitation-learning methods such as behavior cloning and ACT. MolmoSpaces may be explored as an extension for greater object and geometry diversity. Relevant work on part-aware manipulation, visual representation learning, and generalizable robot policies will also guide the model and evaluation design.